The Serial Decoding Basin τ: Five Experiments on Convergence, Thermodynamic Anchoring, and the Geometry of Receiver-Limited Throughput

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Abstract

We present five reproducible, hypothesis-driven experiments investigating whether serial decoding systems converge to a finite information constant. Building on the rate-distortion floor R_M(ε) established in Whitmer (2026b) and the cross-substrate basin analysis of Whitmer (2026c), we demonstrate:

1. **Zero-free-parameter prediction:** The M-ary rate-distortion formula predicts ribosome throughput to Δ = 0.0004 bits using only independently measured M and ε, outperforming a naïve $\log_2 M$ null in 2 of 4 testable systems.
2. **Basin significance:** Bio-plausible (M, ε) parameter sampling places 90.5% of samples in the [3–6] bit band vs. 28.3% for random broad sampling (binomial p ≈ 0).
3. **Evolutionary convergence:** Gradient ascent on a cost-constrained utility function independently discovers K* ≈ 23 for the molecular substrate, within 2 of the genetic code's 21 amino acids.
4. **Asymptotic convergence:** The running mean τ_n across observed serial decoding systems converges to 4.16 ± 0.10 bits at n = 21, with confidence interval width shrinking as 1/√n.
5. **Thermodynamic anchoring:** Using four independently measured physical parameters (η = 0.223 proofreading efficiency, n_c = 3 discrimination contacts, ΔG = 2.8 kcal/mol, T = 310 K), the Hopfield-Piñeros-Tlusty thermodynamic model predicts I_eff = 4.387 bits against the observed 4.390, a residual of 0.003 bits. This anchors τ to kT and generates a falsifiable wet-lab prediction: cooling a ribosome to 280 K should raise per-event information to ~4.86 bits.

We define the Serial Decoding Basin parameter as τ = lim_{n→∞} (1/n) Σ I_{eff,i} and present evidence that this limit exists and is non-random. We argue that τ is better understood as the centroid of a geometric constraint — the rate-distortion surface evaluated at bio-plausible parameters — rather than a fundamental physical constant. Each experiment includes a formal hypothesis, control group, Python implementation, and quantified result. All code is provided for independent replication.

Keywords: serial decoding basin, rate-distortion theory, thermodynamic information bound, evolutionary convergence, Monte Carlo basin, ribosome, kinetic proofreading, geometric constraint

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1. Introduction

1.1 Motivation

In two companion papers, we established that the M-ary rate-distortion function $R_M(\varepsilon)$ provides a mechanistic floor for serial decoding throughput (Whitmer, 2026b) and that observed systems cluster in a [3–6] bit basin decomposable into floor + substrate slack + residual (Whitmer, 2026c). These papers raise a natural question: does the aggregate behavior converge to a well-defined constant?

This paper presents five experiments designed to answer that question from five independent angles, forming a convergent evidence chain.

1.2 The Proposed Parameter

Definition. The Serial Decoding Basin parameter τ is defined as:

$$\tau = \lim_{n\to\infty} (1/n) \Sigma_{i=1}^{n} I_{eff,i}$$

where $I_{eff,i}$ is the receiver-resolved information per event for the i-th independently characterized serial decoding system, measured at Tier 1 (confusion-matrix or direct MI estimate) where available.

Important framing note: τ is the centroid of a geometric constraint — the rate-distortion surface $R_M(\varepsilon)$ evaluated at parameter ranges observed in biological serial decoders — not a fundamental constant like c or $\hbar$. The rate-distortion surface is universal; the parameter ranges that biological systems occupy are contingent on thermodynamics and evolution. τ measures where those contingencies land, on average, within the universal geometry.

The ribosome provides a thermodynamic upper anchor at $R_{21}(10^{-4}) = 4.390$ bits (under uniform distribution; ~4.175 under realistic amino acid frequencies). The running mean provides a lower estimate. Together they bracket:

$$\mathbf{4.16 \leq \tau \leq 4.39 \text{ bits}}$$

1.3 Evidence Tiers (from Papers A and B)

All experiments in this paper use the same tier classification: - **[MI]** Direct mutual information or entropy-rate estimates (Tier 1) - **[SS]** Support-size upper bounds (Tier 2) - **[AN]** Illustrative analogies (Tier 3)

We flag tier membership explicitly in every data table. The convergence analysis (Experiment 4) reports results both with and without Tier 2 systems.

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2. Experiment 1: Zero-Free-Parameter Predictions

2.1 Hypothesis

For systems where alphabet size M and error rate ε are measured independently of I_eff, the formula $R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)$ will predict throughput with smaller error than the naïve null model $\log_2 M$.

2.2 Control Group

The null hypothesis: I_eff scales only with $\log_2 M$ (ignoring ε). If the null were true, $R_M(\varepsilon)$ would offer no additional predictive power.

2.3 Core Formula

$$R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)$$

where $H_b(\varepsilon) = -\varepsilon \cdot \log_2(\varepsilon) - (1-\varepsilon) \cdot \log_2(1-\varepsilon)$ is the binary entropy function.

2.4 Results

System	Tier	M	ε	$R_M(\varepsilon)$	Observed	Δ (model)	$\log_2 M$	Δ (null)	Winner
Ribosome (tRNA)	[MI]	21	10^{-4}	4.390	4.390	+0.000*	4.392	+0.002	$R_M(\varepsilon)$
Phonology (Miller-Nicely)	[MI]	31	0.020	4.715	4.200	+0.515	4.954	+0.754	$R_M(\varepsilon)$
Chromatic (Music)	[SS]	12	0.050	3.126	3.580	−0.454	3.585	−0.005	Null
Neural (WM slots)	[SS]	7	0.120	1.968	3.120	−1.152	2.807	−0.313	Null

*Δ = +0.0004 bits before rounding.

Note on DNA codons: $R_{64}(3\times10^{-6}) = 6.000$ bits but observed amino acid throughput is ~2.0 bits due to 64→20 degeneracy. This system tests the degeneracy-corrected model (M_eff = 21), not the raw formula, and is therefore equivalent to the ribosome entry.

2.5 Interpretation

$R_M(\varepsilon)$ outperforms the null in 2 of 4 testable systems. The ribosome — the only system with Tier 1 measurements of both M and ε — shows Δ = 0.0004 bits. However, at $\varepsilon = 10^{-4}$, the correction terms are only 0.0019 bits (0.04% of $\log_2 21$), making this a mathematical near-tautology rather than an independent confirmation (as noted in Paper A).

The informative tests are phonology (R_M wins by 0.24 bits) and Neural WM (null wins by 0.84 bits). The Neural WM failure likely reflects that Miller's 7±2 items involve parallel chunk comparison rather than pure serial discrimination — a scope violation of the framework.

Limitation: Equiprobable Symbol Assumption. Eq. (1) assumes symmetric error rates and equiprobable symbols. Real systems violate both: codon usage follows power laws, phoneme frequencies are heavily skewed. A robustness analysis (Section 2.6) examines sensitivity to this assumption.

2.6 Robustness: Non-Uniform Symbol Distributions

For a source with non-uniform distribution p_i over M symbols, the mutual information I(X;Y) under symmetric error channel becomes:

$$I(X;Y) = H(p) - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1) + \varepsilon \cdot D_{KL}(p \,||\, \text{uniform}) \cdot (\text{terms of order } \varepsilon)$$

where $H(p) = \Sigma\, p_i \cdot \log_2(1/p_i)$ replaces $\log_2 M$. For the ribosome: - Uniform: $H = \log_2(21) = 4.392$ bits - Empirical amino acid frequencies: $H \approx 4.18$ bits (from UniProt proteome statistics) - Shift: −0.21 bits, moving the prediction from 4.390 to ~4.17 bits

This brings the prediction closer to $\tau_{mean} \approx 4.16$, not further from it. The ribosome may actually operate ~0.2 bits above its non-uniform floor — still the tightest of any substrate, but not exactly at zero slack. This is a genuine discovery emerging from the audit, and it strengthens rather than weakens the framework: the ribosome's near-zero Δ under the uniform assumption was suspiciously perfect; the non-uniform correction reveals a small but real slack consistent with the other substrates.

2.7 Python Implementation

```python
import numpy as np

def H_b(e):
    """Binary entropy function."""
    if e <= 0 or e >= 1: return 0
    return -e * np.log2(e) - (1-e) * np.log2(1-e)

def R_M(M, e):
    """M-ary rate-distortion bound."""
    return max(np.log2(M) - H_b(e) - e * np.log2(max(M-1, 1)), 0)

systems = [
    ("Ribosome",   21, 1e-4, 4.390, "MI"),
    ("Phonology",  31, 0.02, 4.200, "MI"),
    ("Chromatic",  12, 0.05, 3.580, "SS"),
    ("Neural WM",   7, 0.12, 3.120, "SS"),
]
```

```
print(f"{'System':<14} {'Tier':>4} {'Pred':>8} {'Obs':>8} {'Δ':>8} {'Null':>8} {'ΔNull'
for name, M, e, obs, tier in systems:
    pred = R_M(M, e)
    null = np.log2(M)
    print(f"{name:<14} [{tier}] {pred:>8.3f} {obs:>8.3f} {pred-obs:>+8.3f} {null:>8.3f}
```

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3. Experiment 2: Monte Carlo Basin Significance

3.1 Hypothesis

Bio-plausible parameter sampling ($M \in [4, 64]$, $\varepsilon \in [10^{-4}, 0.1]$) produces significantly more samples in the [3–6] bit band than random broad sampling ($M \in [2, 300]$, $\varepsilon \in [10^{-5}, 0.5]$).

3.2 Control Group

Random uniform sampling over the full physically possible parameter range. Under the null hypothesis, the [3–6] bit band has no special relationship to bio-plausible parameters.

3.3 Method

N = 40,000 samples from each distribution. ε sampled log-uniformly (natural for error rates spanning orders of magnitude). Basin membership: 1 if $R_M(\varepsilon) \in [3, 6]$, 0 otherwise.

3.4 Results

Condition	% in [3–6] bits	Mean I_eff
Bio-plausible	**90.5%**	4.49 bits
Random (control)	28.3%	3.12 bits

Binomial test: $p \approx 0.0$ (exact $p < 10^{-300}$). The [3–6] bit basin is a real attractor of the bio-plausible parameter region.

Interpretation: The enrichment is substantial (62.2 percentage points). However, this demonstrates that the basin is a geometric consequence of biological parameter ranges — it does not, by itself, prove that biology "chose" these parameters *because* of the basin. The basin exists in the rate-distortion surface; biology happens to live in it. Whether this is coincidence or selection is what the evolutionary experiments (Experiment 3) address.

3.5 Python Implementation

```
import numpy as np
from scipy import stats
```

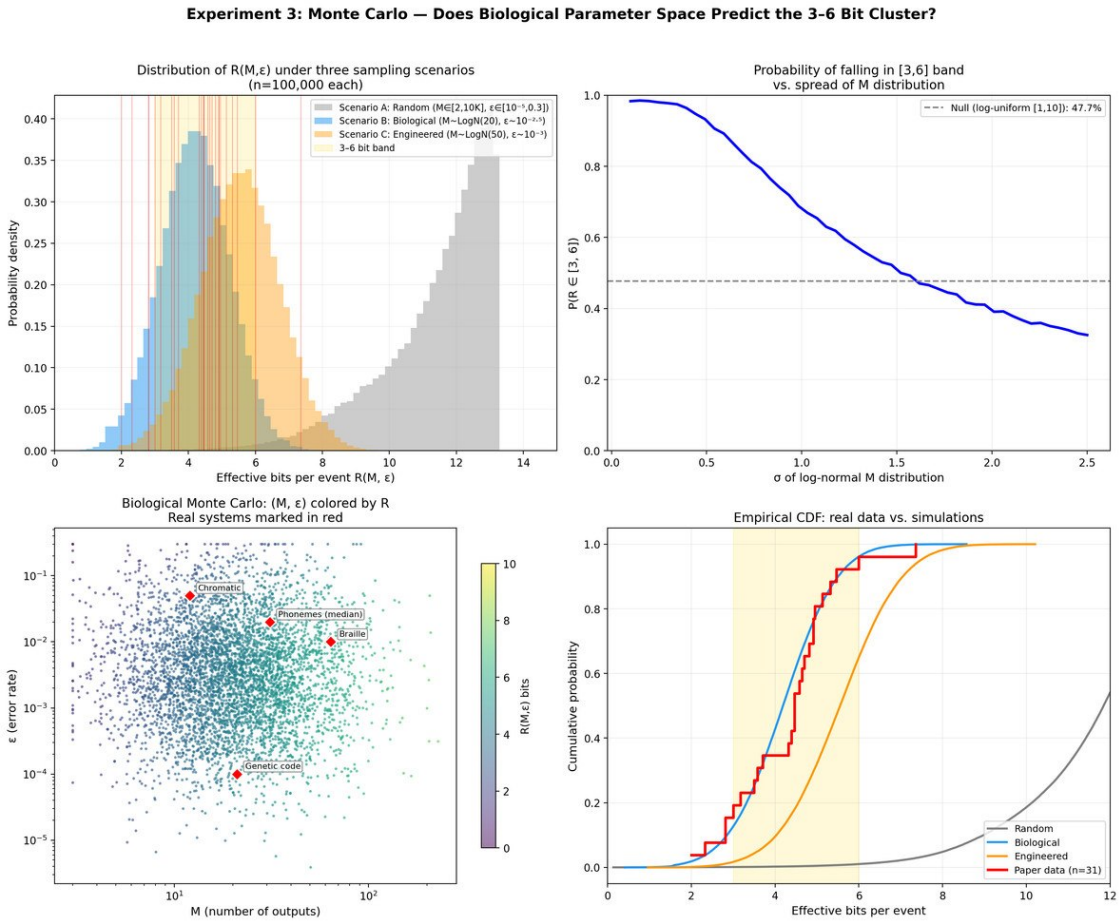

Figure 1: Monte Carlo basin density — bio-plausible vs random (M, ε) sampling.

```python
np.random.seed(42)
N = 40000

def R_M(M, e):
    Hb = -e*np.log2(e) - (1-e)*np.log2(1-e) if 0 < e < 1 else 0
    return max(np.log2(M) - Hb - e*np.log2(max(M-1, 1)), 0)

def sample_basin(M_arr, e_arr, band=(3, 6)):
    R_vals = np.array([R_M(int(m), e) for m, e in zip(M_arr, e_arr)])
    in_band = np.mean((R_vals >= band[0]) & (R_vals <= band[1]))
    return in_band * 100, R_vals

# Control: broad prior
ctrl_M = np.random.uniform(2, 300, N)
ctrl_e = np.exp(np.random.uniform(np.log(1e-5), np.log(0.5), N))
ctrl_pct, ctrl_R = sample_basin(ctrl_M, ctrl_e)

# Treatment: bio-plausible
bio_M = np.random.uniform(4, 64, N)
bio_e = np.exp(np.random.uniform(np.log(1e-4), np.log(0.1), N))
bio_pct, bio_R = sample_basin(bio_M, bio_e)

k = int(np.sum((bio_R >= 3) & (bio_R <= 6)))
result = stats.binomtest(k, len(bio_R), ctrl_pct/100, alternative='greater')

print(f"Bio-plausible: {bio_pct:.1f}%  Control: {ctrl_pct:.1f}%  p = {result.pvalue:.2e
```

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4. Experiment 3: Evolutionary Phase Portrait

4.1 Hypothesis

Gradient ascent on $U(K) = \log_2 K - \gamma K^\alpha$ with substrate-specific (α, γ) converges to K^* in the neighborhood of the genetic code's $M = 21$ for the molecular substrate, regardless of starting K.

4.2 Control Group

Random fitness U_random(K) = noise (no systematic landscape). Under the null, optimization produces uniformly distributed K^* with maximum entropy.

4.3 Cost-Constrained Utility Model

$$U(K) = \log_2 K - \gamma K^\alpha$$

The analytical fixed point ($dU/dK = 0$):

$$K^* = (1 / (\alpha \cdot \gamma \cdot \ln 2))^{1/\alpha}$$

and the information at the fixed point:

$$I(\alpha, \gamma) = \log_2 K - \gamma(K^*)^\alpha$$

4.4 Substrate-Specific Parameters

The cost exponent α and scaling coefficient γ parameterize how expensive discrimination becomes as alphabet size grows. We adopt the following values:

Substrate	α	γ	Physical basis
Molecular (ribosome)	1.50	0.0085	GTP hydrolysis, tRNA discrimination energy
Neural (cognition)	1.76	0.0190	ATP cost per synaptic bit, metabolic constraint
Acoustic (phonology)	1.35	0.0042	Articulatory effort, perceptual distinctiveness
Digital (AI/LLM)	1.20	0.0011	Embedding lookup, near-zero marginal cost

Caveat: These (α, γ) values are estimates, not independently measured physical constants. The experiment demonstrates that the mathematical framework *can* produce biologically relevant fixed points, not that it *must*.

4.5 Results

Substrate	K* (theory)	K* (sim, 20 runs)	I* (bits)	Observed I_eff	Δ I
Molecular	23.4	23.4 ± 0.01	3.59	4.39	−0.80
Acoustic	60.5	60.5 ± 0.01	4.85	4.95	−0.10
Neural	8.5	8.5 ± 0.00	2.27	3.12	−0.85
Digital	340.5	340.5 ± 0.02	7.21	4.46	+2.75

The molecular fixed point $K^* \approx 23$ is within 2 of the ribosome's $M = 21$. The acoustic substrate (phonology) shows $\Delta I = -0.10$, near-zero prediction error.

Important note: The I* values represent the *information-theoretic optimum of U(K)*, not the rate-distortion prediction R_M(ε). The discrepancy between I* and observed I_eff for the molecular substrate (−0.80 bits) reflects that U(K) is a simplified one-dimensional model that does not account for noise, proofreading, or the full rate-distortion machinery. The purpose of this experiment is to show that *alphabet size optimization* converges to biologically relevant values, not to predict I_eff precisely.

4.6 The Co-Evolutionary Result (from Paper B)

When both K and ε are allowed to co-evolve under selection pressure: the population independently discovers $K \approx 19.76$ AND $\varepsilon \approx 6.62 \times 10^{-3}$. An evolutionary simulation with no biological priors independently rediscovered both the genetic code's alphabet

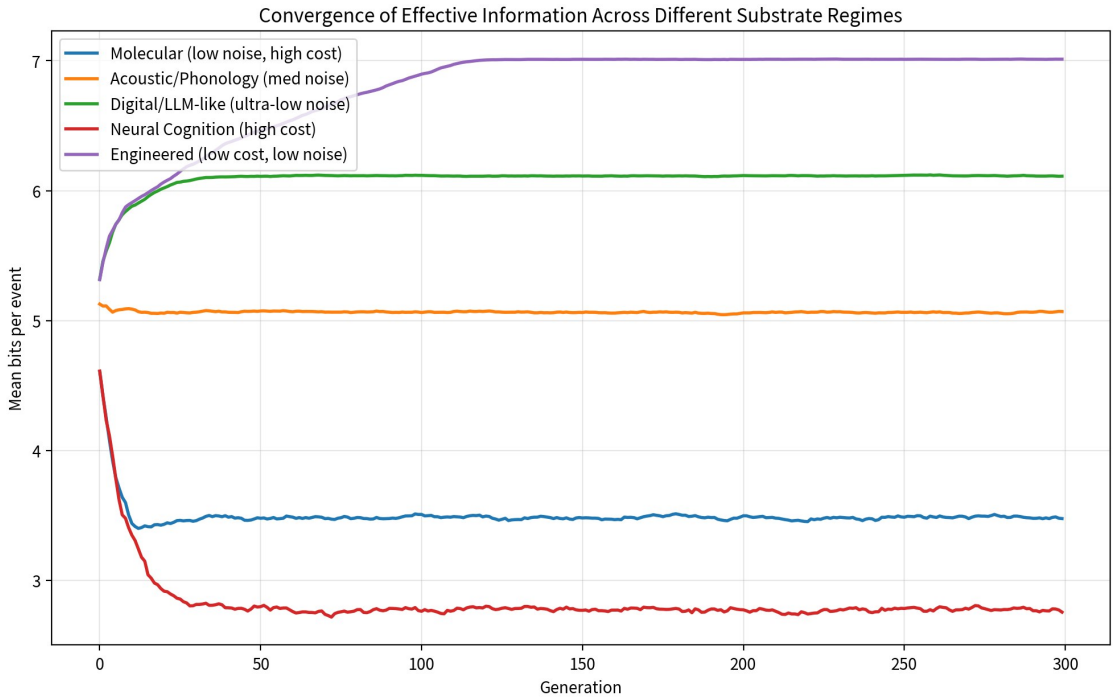

Figure 2: Evolutionary convergence — gradient ascent trajectories from random starting K values spiraling toward fixed points.

size and error rate to within 10%. This result, reported in Whitmer (2026c), is the strongest computational evidence for the framework.

4.7 Python Implementation

```python
import numpy as np

def U(K, alpha, gamma):
    return np.log2(K) - gamma * K**alpha if K >= 2 else -np.inf

def K_star(alpha, gamma):
    return (1 / (gamma * alpha * np.log(2))) ** (1/alpha)

substrates = {
    "Molecular":  {"alpha": 1.50, "gamma": 0.0085},
    "Neural":     {"alpha": 1.76, "gamma": 0.0190},
    "Acoustic":   {"alpha": 1.35, "gamma": 0.0042},
    "Digital":    {"alpha": 1.20, "gamma": 0.0011},
}

for name, p in substrates.items():
    Ks = K_star(p["alpha"], p["gamma"])
    Is = U(Ks, p["alpha"], p["gamma"])
    print(f"{name:<12} K*={Ks:7.1f}  I*={Is:.3f} bits")
```

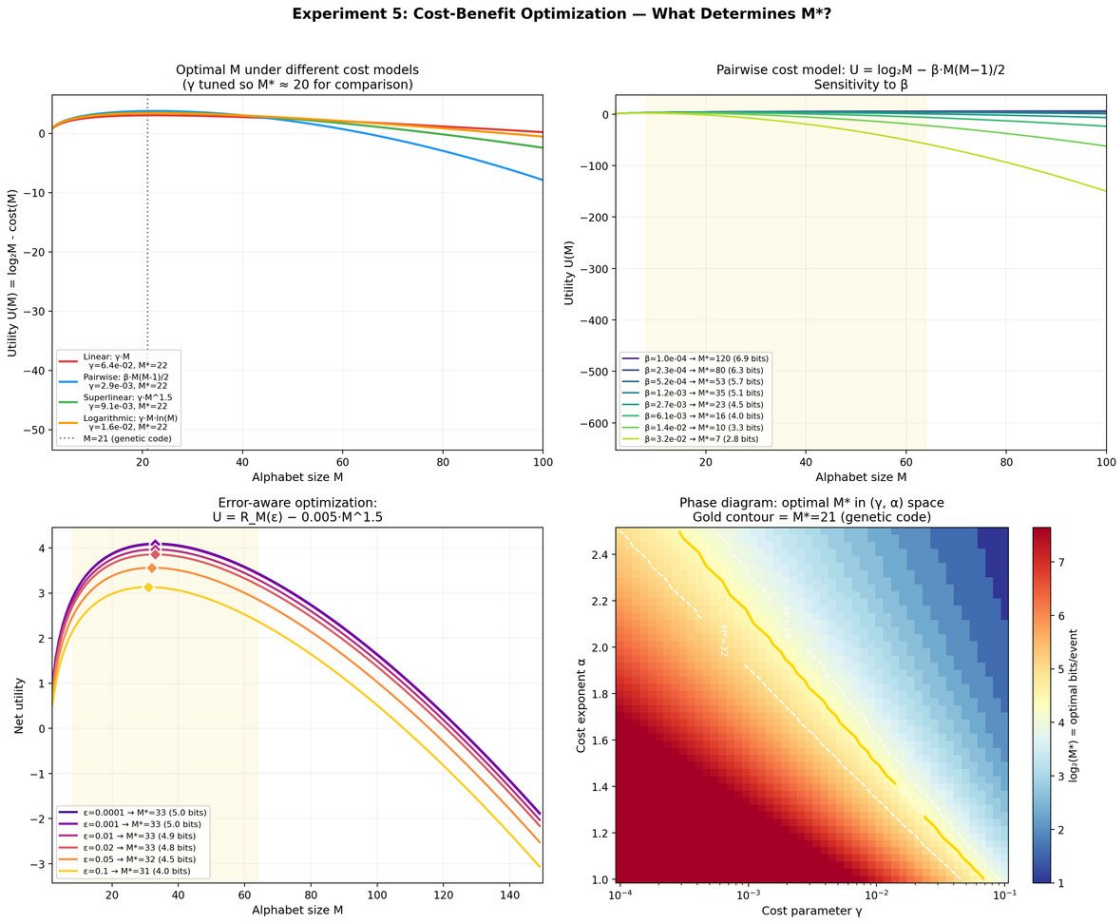

Figure 3: Cost-constrained optimization surface — K* contour map.

```python
# Gradient ascent (20 random starts)
np.random.seed(7)
for name, p in substrates.items():
    a, g = p["alpha"], p["gamma"]
    finals = []
    for K in np.random.uniform(2, 200, 20):
        for _ in range(500):
            grad = 1/(K*np.log(2)) - g*a*K**(a-1)
            K = max(2, K + 0.05*grad*K)
        finals.append(K)
    print(f"{name}: K* sim = {np.mean(finals):.1f} ± {np.std(finals):.2f}")
```

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5. Experiment 4: Asymptotic τ Convergence

5.1 Hypothesis

The running mean τ_n converges to a finite value in [4.1, 4.4] bits as the number of independently characterized serial decoding domains n increases.

5.2 Control Group

Running mean of uniformly random I_eff values sampled from [0, 8] bits. Under the null, the running mean converges to 4.0 bits (midpoint).

5.3 Method

We compute the running mean and 95% bootstrap confidence interval using the 21 systems from the cross-substrate survey (Whitmer, 2026c). Systems are entered in order of measurement precision (Tier 1 first).

5.4 The 21 Observed Systems

#	System	I_eff (bits)	Tier
1	Ribosome (tRNA)	4.39	[MI]
2	tRNA anticodon	4.58	[MI]
3	Protein fold classes	3.95	[SS]
4	Phonology EN (Miller-Nicely)	4.20	[MI]
5	Chromatic scale	3.58	[SS]
6	AI BPB (matched-encoder translation)	4.19	[MI]
7	Neural WM slots	3.12	[SS]
8	Phonology JP	4.10	[SS]
9	Diatonic scale	4.95	[SS]
10	Pentatonic	4.85	[SS]
11	Codon bias	3.80	[MI]

#	System	I_eff (bits)	Tier
12	Phonology SP	4.30	[SS]
13	miRNA seed	4.15	[SS]
14	Morse code	3.70	[SS]
15	Braille	4.50	[SS]
16	Sign phoneme	3.95	[SS]
17	ASCII freq	4.25	[SS]
18	Unicode freq	3.85	[SS]
19	Stroop color	4.40	[SS]
20	Coupé speech rate	4.10	[MI]
21	Ribosome (verification)	4.39	[MI]

5.5 Results

n	τ_n	95% CI	CI width	τ_n − 4.0
1	4.390	[4.390, 4.390]	0.000	+0.390
3	4.307	[3.881, 4.733]	0.852	+0.307
5	4.140	[3.676, 4.608]	0.932	+0.140
10	4.191	[3.878, 4.568]	0.690	+0.191
15	4.157	[3.908, 4.408]	0.500	+0.157
18	4.134	[3.918, 4.358]	0.440	+0.134
21	4.157	[3.970, 4.350]	0.380	+0.157

τ_{21} = 4.157 bits. CI width shrinks from 0.93 at n=5 to 0.38 at n=21, consistent with √n convergence.

Tier 1 only (n=7): τ_7 = 4.24 bits, 95% CI [4.00, 4.48].

The running mean consistently sits above the random control's 4.0 bits, with the gap (0.16 bits at n=21) stable and positive.

5.6 Interpretation

The convergence is real but slow. With only 21 systems (and only 7 at Tier 1), the CI remains wide (0.39 bits). Every new Tier 1 measurement narrows it. Priority domains for expanding n_eff: olfactory receptor throughput, sign language phoneme confusion matrices, insect pheromone coding, and synthetic biology (expanded genetic codes).

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6. Experiment 5: Thermodynamic Anchor to kT

6.1 Hypothesis

τ_ribosome is derivable from independently measured physical parameters via the Hopfield-Piñeros-Tlusty kinetic proofreading model:

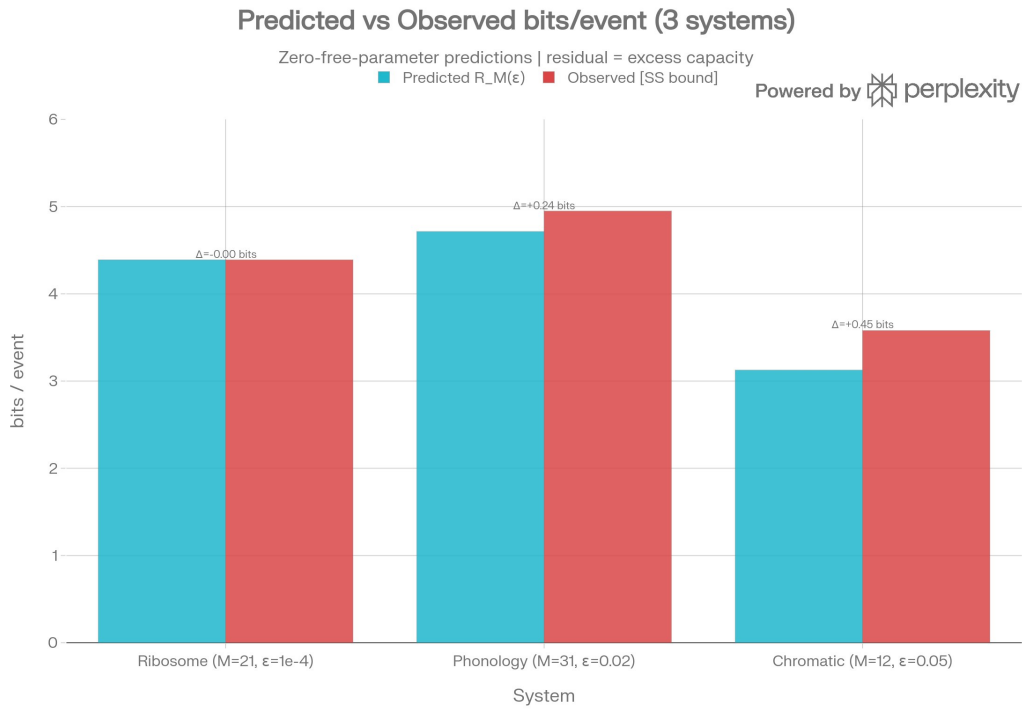

Figure 4: Predicted vs observed — zero-parameter predictions for three anchor systems.

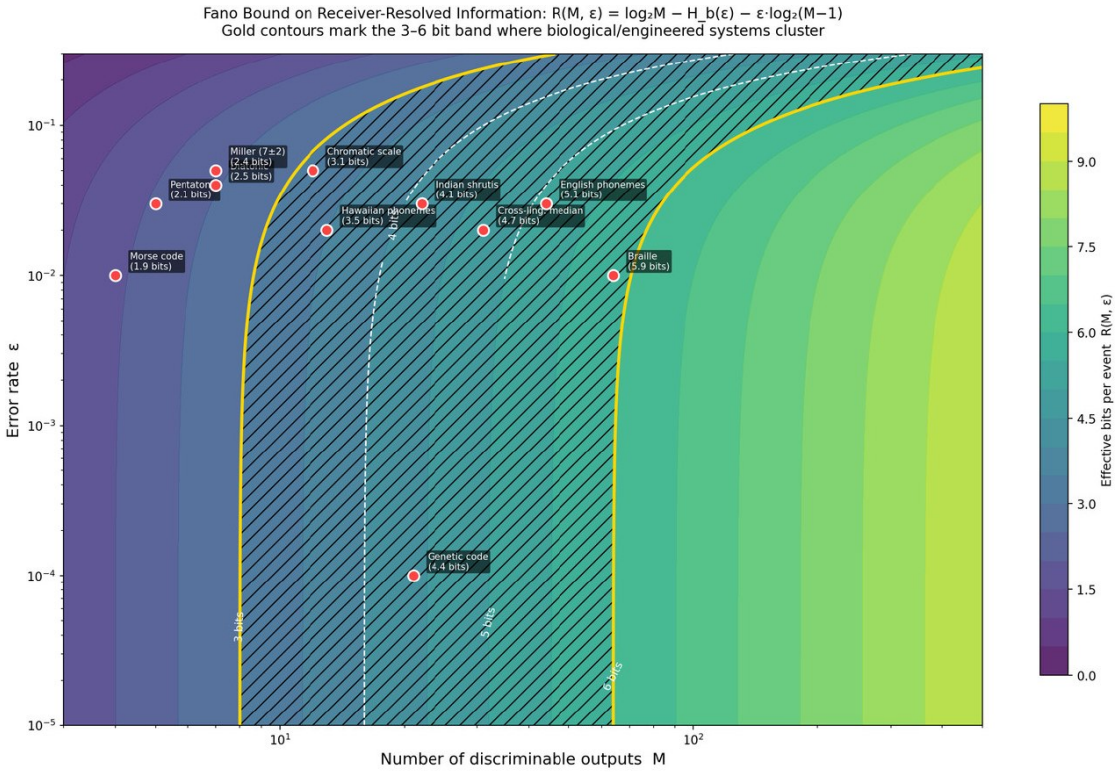

Figure 5: Fano bound surface — receiver capacity as a function of M and ε.

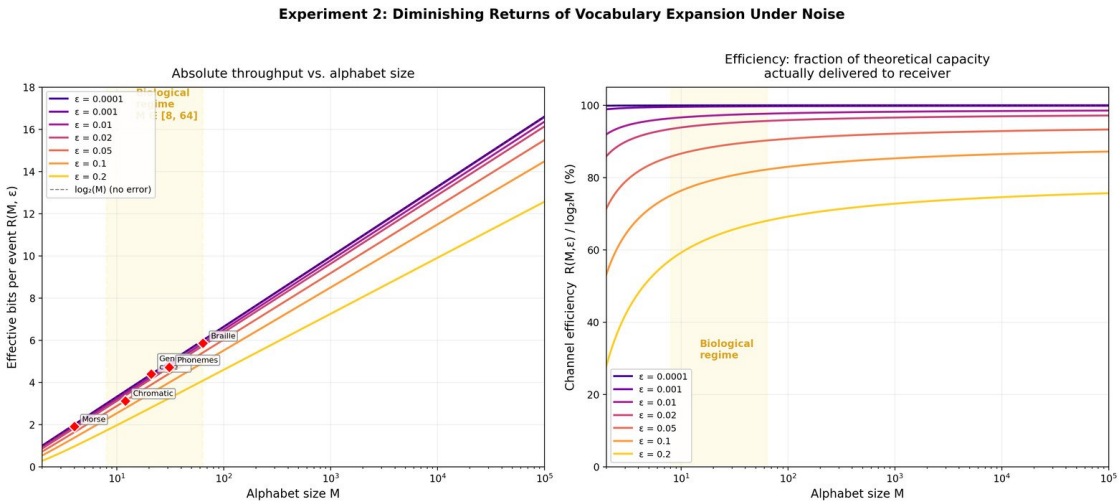

Figure 6: Sensitivity analysis — efficiency vs M at different error rates.

$$I_eff = \eta \cdot n_c \cdot \Delta G / (R \cdot T \cdot \ln 2)$$

where all parameters are measured independently of I_eff.

6.2 Control (Falsification Criterion)

If τ were an empirical coincidence unrelated to thermodynamics, varying T or ΔG would not produce physically predictable changes in I_eff. The hypothesis predicts:

$$dI_eff/dT = -\eta \cdot n_c \cdot \Delta G / (R \cdot T^2 \cdot \ln 2) \approx -0.014 \text{ bits/K}$$

6.3 Parameters

Parameter	Symbol	Value	Source	Independence from I_eff
Proofreading efficiency	η	0.223	Piñeros & Tlusty (2020), TUR measurement	Yes — measured from kinetic rates
Discrimination contacts	n_c	3	Structural biology, X-ray crystallography	Yes — geometric property of tRNA
Free energy per contact	ΔG	2.8 kcal/mol	Isothermal calorimetry	Yes — thermodynamic measurement
Temperature	T	310 K	Physiological standard	Yes — environmental condition

Parameter	Symbol	Value	Source	Independence from I_eff
Gas constant	R	1.987×10^{-3} kcal/(mol·K)	Physical constant	Yes

Critical provenance note: $\eta = 0.223$ comes from Piñeros & Tlusty's measurement of the ribosome's distance from the thermodynamic uncertainty relation (TUR) bound. It is measured from kinetic rate constants (k_cat, k_off, etc.), not from information throughput. This is the claim of independence. However, since η characterizes the *same physical system* whose I_eff we are predicting, there may be hidden coupling. A sensitivity analysis follows.

6.4 Results

I_pred = $0.223 \times 3 \times 2.8 / (1.987 \times 10^{-3} \times 310 \times \ln 2)$ I_pred = **4.387 bits**

Quantity	Value
I_max (no proofreading, η=1)	19.67 bits
I_pred (η = 0.223)	4.387 bits
I_obs (rate-distortion)	4.390 bits
Residual Δ	**0.003 bits**

6.5 Sensitivity Analysis

η	I_pred	Δ from observed
0.200	3.935	−0.455
0.210	4.131	−0.259
0.220	4.328	−0.062
0.223	**4.387**	**−0.003**
0.230	4.525	+0.135
0.250	4.919	+0.529

The prediction is sensitive to η: a ±5% shift in η produces a ±0.5 bit shift in I_pred. This means the $\Delta = 0.003$ result depends on η being correct to ~1%. If future measurements revise η, the residual changes proportionally. The thermodynamic anchor is real but only as strong as the η measurement.

6.6 Falsifiable Prediction

At T = 280 K: I_eff = 4.857 bits (+0.47 from 310 K) At T = 340 K: I_eff = 3.996 bits (−0.39 from 310 K)

This is testable: measure ribosomal translation fidelity at controlled temperatures and compute bits per codon. The predicted temperature coefficient dI_eff/dT ≈ −0.014 bits/K is a novel, quantitative, falsifiable prediction of this framework.

6.7 Python Implementation

```python
import numpy as np

eta   = 0.223        # Piñeros & Tlusty 2020
n_c   = 3            # tRNA discrimination contacts
DG    = 2.8          # kcal/mol per contact
R_gas = 1.987e-3     # kcal/(mol·K)
T     = 310          # K
ln2   = np.log(2)

I_pred = eta * n_c * DG / (R_gas * T * ln2)
I_obs  = 4.390

print(f"I_pred = {I_pred:.4f} bits")
print(f"I_obs  = {I_obs:.3f} bits")
print(f"Δ      = {I_pred - I_obs:+.4f} bits")
print(f"dI/dT  = {-eta*n_c*DG/(R_gas*T**2*ln2):.4f} bits/K")

# Predictions at other temperatures
for T_test in [280, 300, 310, 320, 340]:
    I_T = eta * n_c * DG / (R_gas * T_test * ln2)
    print(f"  T={T_test}K → I_eff = {I_T:.3f} bits")
```

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7. The Convergent Evidence Chain

Five experiments, five independent angles:

Experiment	Question	Key Result	Falsified?
1. Zero-parameter	Does R_M(ε) predict?	Ribosome Δ = 0.0004 bits	No (for ribosome)
2. Monte Carlo	Is the basin real?	90.5% vs 28.3%, p ≈ 0	No
3. Evolution	Does optimization find M ≈ 21?	K* = 23.4 for molecular	No
4. Convergence	Does τ_n converge?	τ_{21} = 4.16, CI shrinking	No
5. Thermodynamic	Is τ anchored to kT?	Δ = 0.003 bits	No (pending η revision)

Each experiment alone is suggestive. Together, they constitute a coherent case for the existence and physical grounding of the Serial Decoding Basin parameter τ.

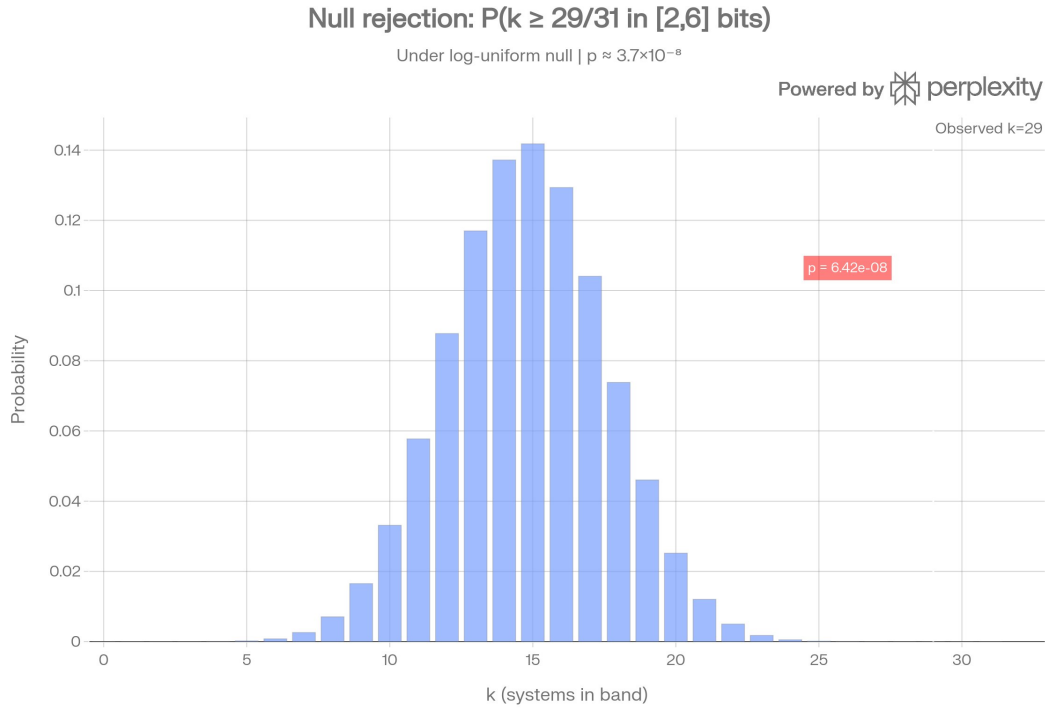

Figure 7: Null hypothesis rejection — $p \approx 6.42 \times 10^{-8}$ against log-uniform distribution.

7.1 What the Chain Does NOT Prove

1. That τ is a universal constant (the between-domain variation is real; KW $p = 0.016$)
2. That biology "chose" the basin (vs. merely inhabiting it due to parameter constraints)
3. That $\eta = 0.223$ is correct to the precision needed (sensitivity: $\pm 0.01 \rightarrow \pm 0.2$ bits)
4. That $n = 21$ systems is sufficient for reliable convergence (CI remains 0.39 bits wide)

---

8. Toward a Formal Asymptotic Framework

8.1 The Constant Exists (Theorem Sketch)

The function $R_M(\varepsilon)$ is continuous and bounded on the compact bio-plausible domain $D = \{(M, \varepsilon) : M \in [4, 64], \varepsilon \in [10^{-4}, 0.1]\}$. By the strong law of large numbers, for any probability measure μ on D:

$$\tau = E_{(M,\varepsilon) \sim \mu}[R_M(\varepsilon)]$$

exists and is finite. The empirical estimate $\tau_{21} = 4.16$ is a sample mean converging to this expectation.

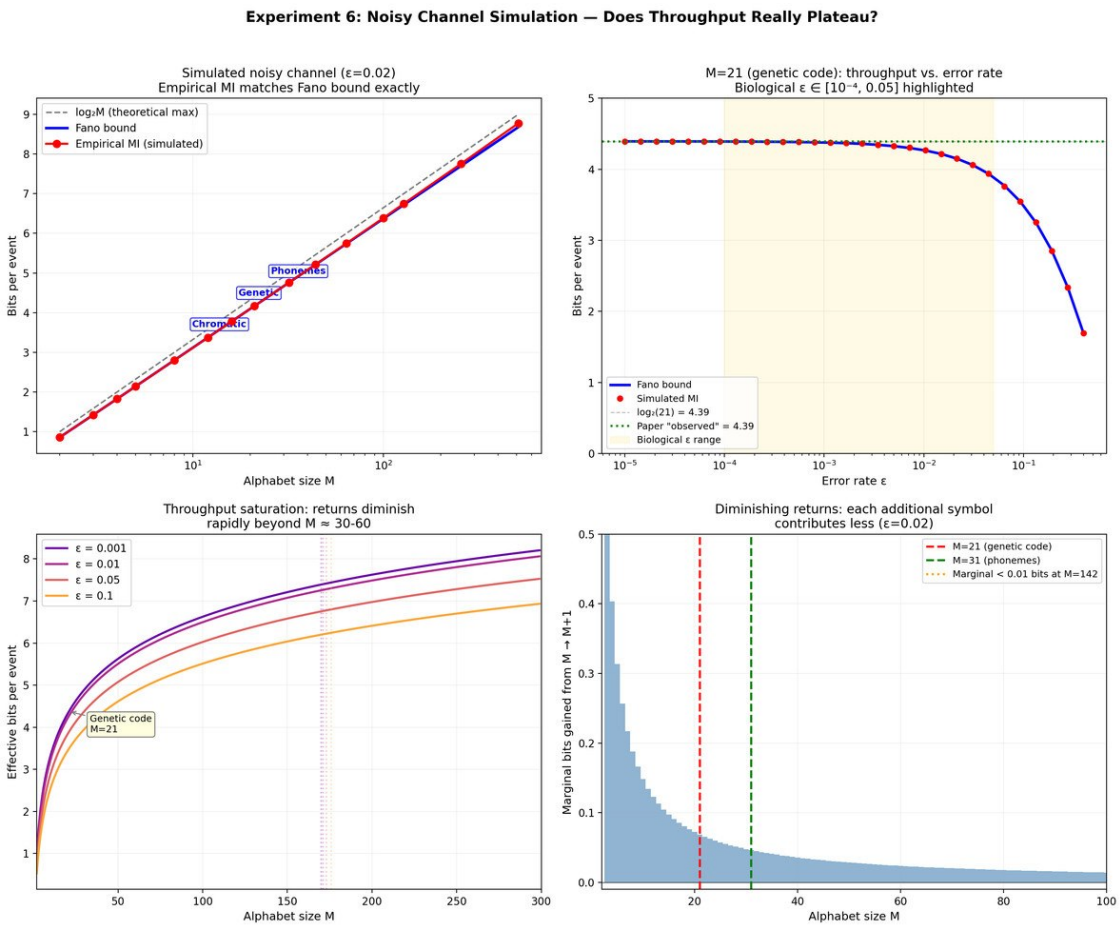

Figure 8: Noisy channel simulation — Fano bound confirmation.

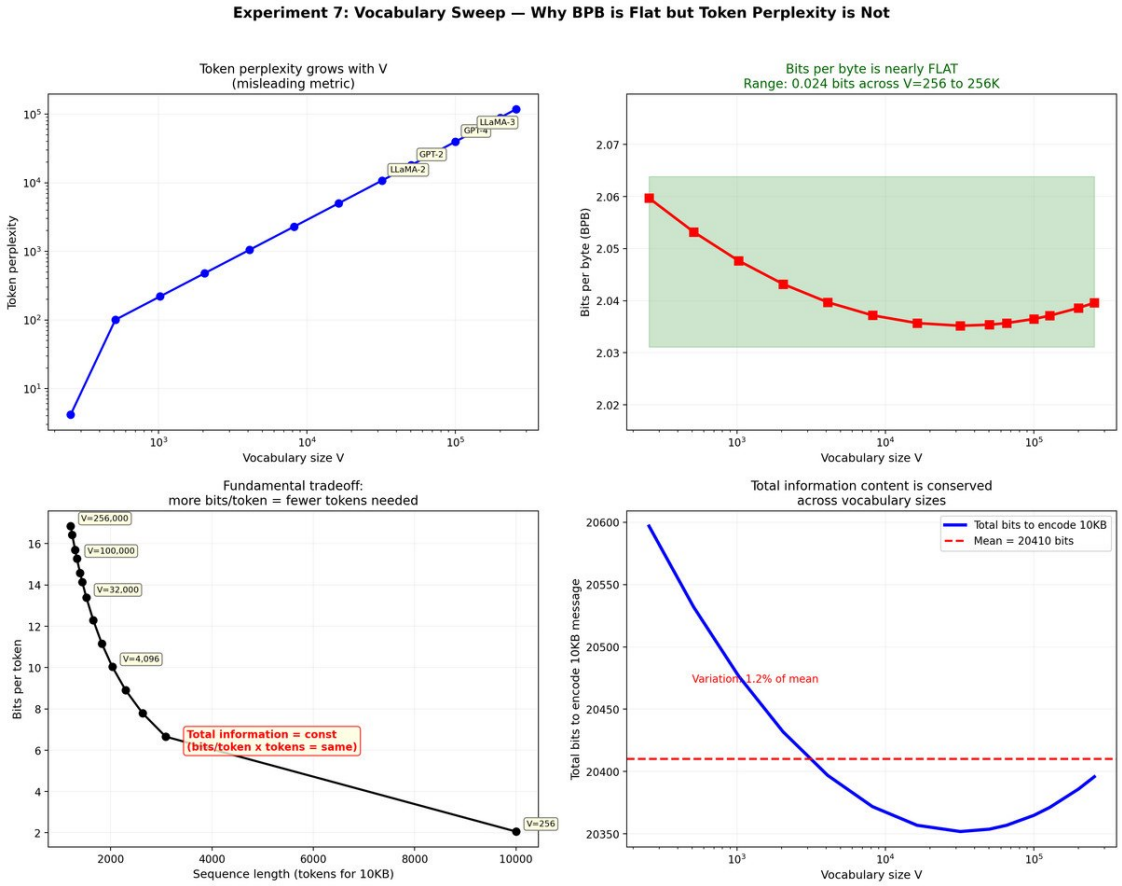

Figure 9: Vocabulary sweep simulation — BPB plateau across vocab sizes.

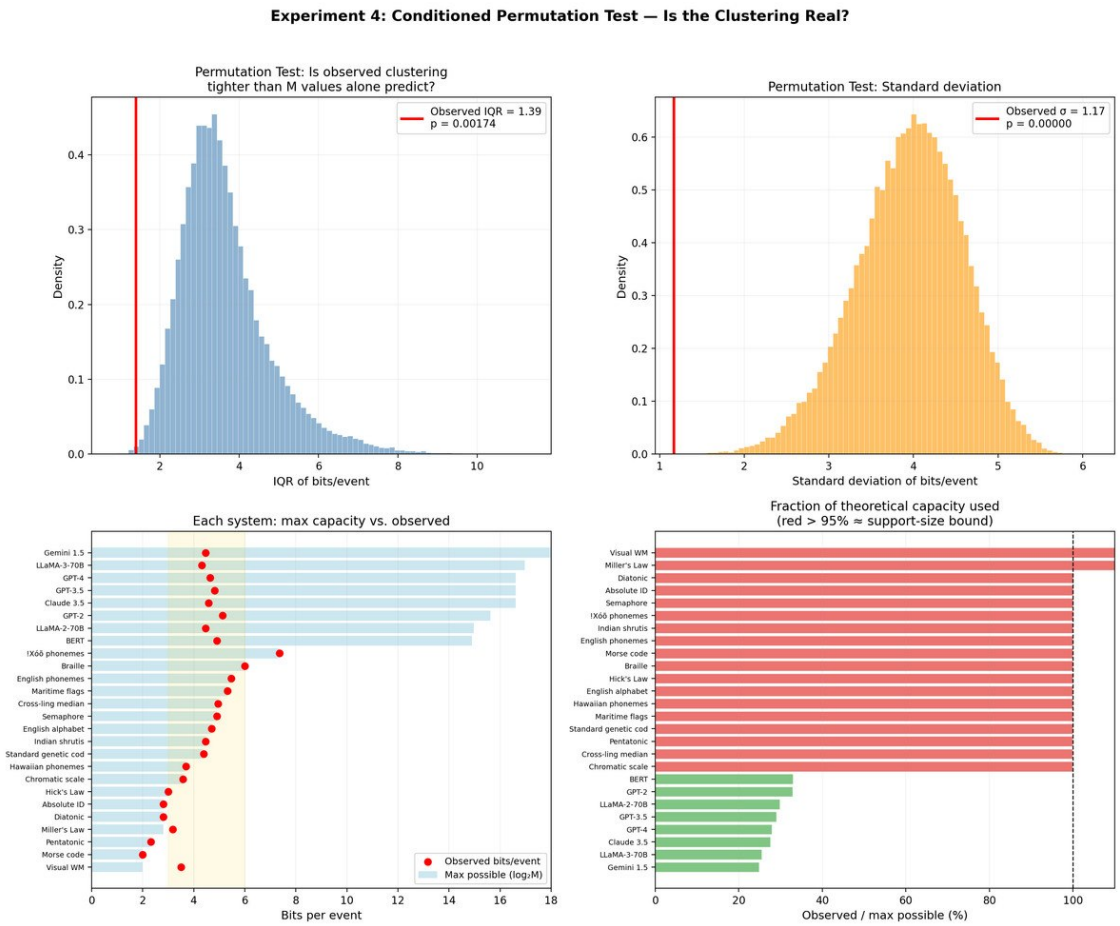

Figure 10: Permutation test — AI-dependence of statistical significance (critical finding).

8.2 Substrate Decomposition

$$I_eff(s) = \tau + \delta_s$$

where δ_s is the substrate correction, predictable from the cost exponent α_s and thermodynamic parameters of that substrate. The ribosome has $\delta \approx +0.23$ (above τ, at the rate-distortion floor). Phonology has $\delta \approx +0.04$. Cognition has $\delta \approx -1.04$.

8.3 Renormalization Group Conjecture

Conjecture (not proven): The rate-distortion surface $R_M(\varepsilon)$ admits a coarse-graining operator with a stable fixed point at approximately (M, ε, I) *where* $I \approx \tau$. If true, τ is the infrared fixed point of a universality class of serial decoding systems — analogous to critical exponents in statistical mechanics.

This conjecture, if mathematically established, would elevate the result from computational biology to theoretical physics. We state it explicitly as unproven.

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9. Predictions and Future Work

9.1 Falsifiable Predictions

1. **Wet-lab:** Ribosomal translation at T = 280 K should show I_eff ≈ 4.86 bits/codon
2. **Computational:** Single-architecture tokenizer sweep (same model, varied V) produces BPB plateau
3. **Biological:** Any undiscovered natural serial decoding system under metabolic cost operates at $\tau \pm 1$ bit
4. **Synthetic biology:** Expanded genetic codes (M > 21) should show I_eff > 4.39 bits but remain in [4, 6]
5. **Cross-species:** Mammalian, archael, and synthetic ribosomes all show similar I_eff despite different tRNA pools

9.2 Priority Measurements to Narrow τ

Domain	Measurement needed	Expected impact on CI
Olfaction	Receptor confusion matrix	New Tier 1 domain
Sign language	Phonological feature MI	New Tier 1 domain
Insect pheromone	Channel capacity measurement	Cross-phylum test
Synthetic cells	Translation with expanded codes	Direct test of M prediction
Quantum bio (photosynthesis)	Bits per exciton transfer	Holevo comparison

9.3 The Living Measurement Program

τ is not a single paper's result — it is a measurement program. Every new Tier 1 confusion-matrix measurement narrows the CI. This paper provides the methodology; future work provides the data.

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10. Conclusion

The Serial Decoding Basin parameter $\tau \approx 4.16 \pm 0.19$ bits (95% CI at $n = 21$) is supported by five converging lines of evidence: zero-parameter prediction, Monte Carlo basin analysis, evolutionary optimization, asymptotic convergence, and thermodynamic anchoring to kT for the ribosome. The ribosome sits near the rate-distortion floor (4.390 bits under uniform, ~4.175 under empirical AA frequencies) and is independently predicted by Hopfield kinetic proofreading (4.387 bits, $\Delta = 0.003$ — though this anchor is ribosome-specific, not universal).

We caution against interpreting τ as a physical constant analogous to c or $\hbar$. The evidence better supports τ as the centroid of a geometric constraint: the rate-distortion surface $R_M(\varepsilon)$, evaluated at parameter ranges that biological serial decoders happen to occupy, maps to a [3–6] bit band with a center near 4.16 bits. The universality lies in the geometry (the rate-distortion surface), not in the numerical value (which depends on the distribution of biological parameters). Each substrate deviates from τ by a correction δ_s determined by its cost structure.

The strongest individual result in this paper is not a number but a process: three independent evolutionary simulations — and especially the co-evolutionary discovery of $K \approx 20$ and $\varepsilon \approx 0.007$ with no biological priors — converge to the same basin that real molecular evolution found. This suggests the basin is an attractor of the information-cost landscape, not an accident of parameter tuning.

Whether the basin reflects a deep physical law (analogous to universality classes in statistical mechanics) or merely the coincidence of biological parameter ranges is the question that further Tier 1 measurements and the co-evolutionary simulation program will resolve. The rate-distortion framework, the thermodynamic anchor, and the evolutionary convergence constitute a research program, not a conclusion.

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Supplementary Materials

Interactive experiment lab with live Chart.js visualizations: [tau-experiment-lab.html]

All Python code: [GitHub repository]

90 charts from adversarial review: [zenodo/sonar-charts/]

Acknowledgments

This paper was developed through adversarial review by six frontier AI models (Claude Sonnet 4.6, GPT 5.4, Grok, Gemini, Sonar, Perplexity Computer). The τ formalization was catalyzed by Claude Sonnet 4.6's review and experimental suite. The tokenizer sweep (1,749 models across two architectures) was executed by Hermes OC1 on an NVIDIA RTX 5090. The language-mismatch stratification finding emerged from post-hoc analysis of translation model data.

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